

Type 2 diabetes mellitus and obesity in sub-Saharan Africa

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Summary

While communicable diseases such as human immunodeficiency virus/acquired immune deficiency syndrome, malaria, and tuberculosis have continued to pose greater threats to the public health system in sub-Saharan Africa (SSA), it is now apparent that non-communicable diseases such as diabetes mellitus are undoubtedly adding to the multiple burdens the peoples in this region suffer. Type 2 diabetes mellitus (T2DM) is the most common form of diabetes (90–95%), exhibiting an alarming prevalence among peoples of this region. Its main risk factors include obesity, rapid urbanization, physical inactivity, ageing, nutrition transitions, and socioeconomic changes. Patients in sub-Saharan Africa also show manifestations of β -cell dysfunction and insulin resistance. However, because of strained economic resources and a poor health care system, most of the patients are diagnosed only after they have overt symptoms and complications. Microvascular complications are the most prevalent, but metabolic disorders and acute infections cause significant mortality. The high cost of treatment of T2DM and its comorbidities, the increasing prevalence of its risk factors, and the gaps in health care system necessitate that solutions be planned and implemented urgently. Aggressive actions and positive responses from well-informed governments appear to be needed for the conducive interplay of all forces required to curb the threat of T2DM in sub-Saharan Africa. Despite the varied ethnic and transitional factors and the limited population data on T2DM in sub-Saharan Africa, this review provides an extensive discussion of the literature on the epidemiology, risk factors, pathogenesis, complications, treatment, and care challenges of T2DM in this region. Copyright © 2010 John Wiley & Sons, Ltd.

Keywords sub-Saharan Africa; type 2 diabetes mellitus; epidemiology; risk factors; pathogenesis; care challenges

Introduction

The global trend of rising non-communicable diseases (NCDs) such as diabetes mellitus strongly poses health and socioeconomic threats in the 21st century. Several studies from 1994–2006 estimated that 100–246 million people had diabetes worldwide, with numbers projected to reach 380 million by 2025 [1–5]. A recent compilation by the International Diabetes Federation (IDF), *Diabetes Atlas*, fourth edition (2009), estimates an even higher prevalence of 6.4% (285 million people) for the year 2010 and 7.7% (438 million people) by the year 2030 in the 20–79 age group [6]. The largest increase is projected to occur in developing countries, mainly in Asia and Africa, and may represent as much as 70% of all cases. It is also important to note that of all diabetes cases reported, about 90–95% are of type 2 diabetes mellitus (T2DM) while <10% are of type 1 diabetes mellitus and of other specific types [7].

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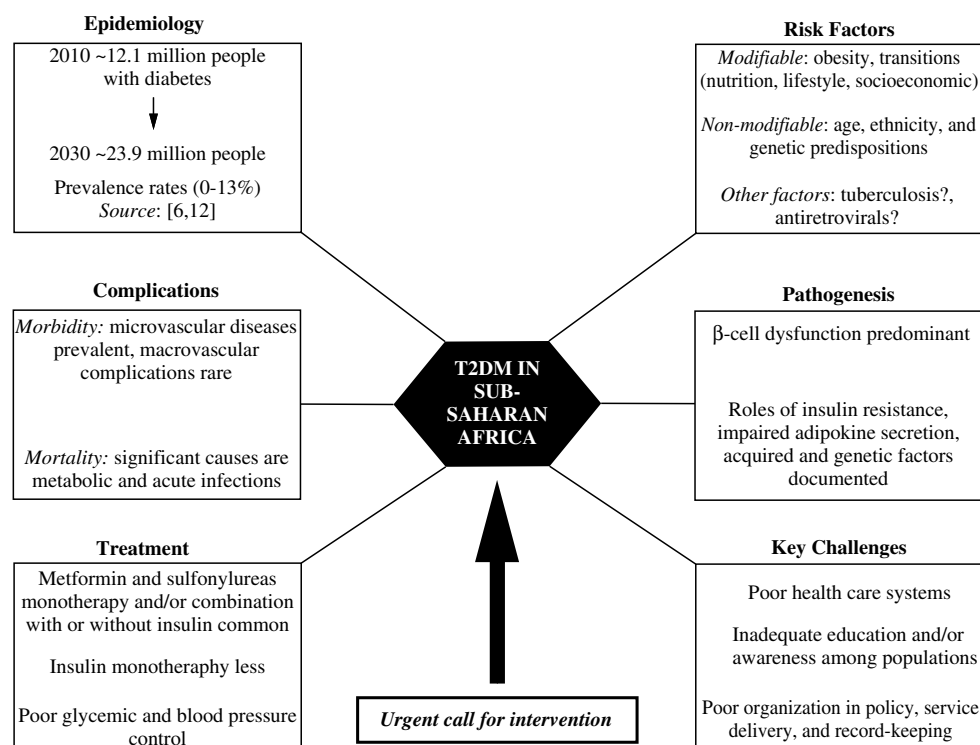


Figure 1. Overview of type 2 diabetes mellitus in sub-Saharan Africa

T2DM is a multiorgan disease that involves both pancreatic β -cell dysfunction and peripheral insulin resistance (IR) in adipose, liver, and muscle cells, of which the diagnosis or overt clinical manifestations may appear after the disease has progressed [8]. Complications characteristic of type 2 diabetic patients include microvascular diseases (e.g. retinopathy, nephropathy, and neuropathy), blindness, slow-healing wounds, hypertension, and cardiovascular disease (CVD), with CVD accounting for the most deaths [9]. Risk factors for the development of T2DM include obesity, ageing, and physical inactivity, often arising as a result of socioeconomic dynamics [7].

While communicable diseases such as human immunodeficiency virus (HIV)/acquired immune deficiency syndrome (AIDS), malaria, and tuberculosis (TB) [10] have continued to pose great challenges in sub-Saharan Africa (SSA), it is now apparent that NCDs such as T2DM are undoubtedly adding to the multiple burdens borne by nations in this region. SSA, with its low-income populace, is also faced with simultaneous underweight and malnutrition problems [11]. The SSA region consists of countries that are fully or partially located south of the Sahara Desert and is largely dominated by the indigenous black people, with some communities of Indian descent, Caucasians, mixed races, and people of Arabic origin. Notwithstanding the varied ethnic and transitional features of this region and the availability of limited data on T2DM in SSA, this review attempts to provide an extensive discussion of the literature on the epidemiology, risk factors, pathogenesis, complications, treatment, and care challenges of T2DM in this region (Figure 1).

Epidemiology

Data on prevalence rates of T2DM in SSA are quite scarce and have mainly included estimates of all types of diabetes. However, T2DM remains the most prevalent, constituting about 90–95% of all cases of diabetes reported [7]. Before the adoption of the World Health Organization (WHO) standardized classification and diabetes diagnostic criteria in 1980 (modified 1985), epidemiological data on diabetes published in SSA, whose detection methods included only urine or blood-based tests, showed prevalence rates of diabetes $\leq 1.4\%$ for all countries except Ivory Coast (5.7%) and South Africa (2.7–3.6%) [12–14]. The rural populations of SSA had lower prevalence rates of 0–0.5% [12–14] when compared to those in urban settings (1–6%) [15,16].

Over the past 24 years, several studies have documented the prevalence of T2DM in SSA by using the WHO standardized methods. These include reports from several different countries in SSA: Western Africa (Cameroon, Ghana, Mauritania, Nigeria, Mali, Guinea, and Togo) [12,17–27], Eastern Africa (Sudan, Tanzania, and Kenya) [12,15,16,28–31], and Southern Africa (South Africa, Zimbabwe, Réunion, Seychelles, and Mauritius) [12,32–39]. Overall, the prevalence rates ranged from 0 (rural Togo) to 10.4% (urban Northern Sudan) [12]. Furthermore, there has been markedly increasing diabetes prevalence in Africa, with the estimate of 3 million in 1994 [1] almost tripling in the next 6 years and reaching 7.1 million in 2000 [40]. The current IDF *Diabetes Atlas* (2009) highlights the increasing problem,

estimating that 12.1 million people in SSA will have diabetes in 2010 (3.8%), with the number projected to double by 2030 [6]. It is noteworthy that data suggest higher prevalence rates in North Africa, which is mainly dominated by people of Arabic and Egyptian ancestry, than most other regions in the expansive SSA [41]. In fact, the highest prevalence in SSA, as noted before in Northern Sudan (10.4%), was among a population of mixed Egyptian ancestry [12]. This trend was ascribed to urbanization; however, genetic factors have not been ruled out. South Africa showed moderate diabetes prevalence rates, which might be due to the higher reported incidences of obesity in the region [10]. Features of T2DM prevalence have been studied globally, and some unique aspects of the epidemic in developing countries have suggested higher prevalence rates in the younger age group, in females, and in urbanized populations [4,42]. Particularly, age-specific data in SSA have indicated an increase in diabetes rates with age; some studies indicate peak prevalence in the 45–64 age group [12,21,28,35,43], whereas other studies indicated peak rates in those older than 65 [12,15,17,23,29,32,36], as characterized in developed nations [42]. With the ravages of HIV/AIDS and malnutrition in SSA, the life expectancy has been reduced to <50 years in some SSA countries, and this might reduce the size of groups that are most at risk for T2DM; but at the same time, acquired factors such as obesity and physical inactivity at an early age might shift the incidence rates to younger populations. In regard to gender distribution, the available reports have not articulated a trend in SSA. Interestingly, global estimates for developing countries have shown that more females have diabetes than males, which is the reverse of the trend in developed nations [4]. A male preponderance of diabetes incidence was reported in studies in Sudan, Cameroon, and rural Tanzania [15,22,29], whereas other studies in Mauritania, Mali, Cameroon, Sudan, and South Africa (Durban) showed a higher prevalence in females [17,23,25,37,43]. Equal rates between genders were evidenced in studies in South Africa (Cape Town), Tanzania, and Sudan [12,28,29,32]. If female preponderance exists in SSA, coupled with the prevalence peaks in patient's reproductive years, it is suggested that there may be high rates of intrauterine fetal exposure to diabetes and future metabolic disorders [42,44]. A variation in diabetes prevalence among ethnic communities has also been reported in SSA; the highest prevalence of diabetes has been reported within the Indian community, mainly in Tanzania (7.1–9.1%) [15,45,46] and South Africa (13%) [35,37,38], compared to the lowest rates in indigenous blacks (1.1–5.3%) [12]. The prevalence rates in Caucasians in SSA have been quite similar to those in Europe (6–10%), which are also relatively high [14].

Risk factors for the increasing prevalence

Globalization, as described by Chapman (2009), is 'a process characterized by the growing interdependence of the world's people, involves the integration of economies, culture, technologies, and governance' [47]. As such, its positive impacts in developing countries, such as those in SSA, are the transition from low to high socioeconomic status (SES), low mortality, higher life expectancy, growth of infrastructure, and reduced incidences of communicable diseases [48]. However, globalization is also characterized by a rapid urbanization pattern, a widening of the gap between the rich and poor, the adaptation of the high-calorie 'western diet', and reduced physical activity, which is negatively impacting health status and health care systems. These changes increase the risks for developing NCDs, such as T2DM [49]. Below, we discuss some of the key modifiable risk factors for T2DM in SSA, including obesity, nutrition and lifestyle changes, and socioeconomic transitions, as well as the non-modifiable factors of ageing and ethnicity (Table 1).

Obesity

Obesity is a fast-growing problem that is reaching epidemic proportions worldwide [50]. It occurs as a result of an imbalance between energy intake and expenditure and is characterized by increased body fat stores. Persistent obesity disrupts metabolic processes controlling blood glucose, blood pressure, and lipids [48]. A body mass index (BMI) of $\geq 30 \text{ kg/m}^2$ is the most commonly

Table 1. Risk factors for the increasing prevalence of type 2 diabetes mellitus in sub-Saharan Africa

Modifiable risk factors	
Obesity	
Anthropometric parameters (body mass index, waist circumference, waist-to-hip ratio, abdominal fat)	
Nutrition/lifestyle changes	
Fetal undernourishment	
Stunting	
Cultural preference of an excess body weight	
Transitions	
Nutrition (dietary changes)	
Lifestyle (physical inactivity, smoking ^a , alcohol use ^a)	
Socioeconomic/demographic (urbanization, mechanization, higher levels of incomes and education)	
Non-modifiable risk factors	
Ageing	
Gender ^a	
Ethnicity (indigenous blacks < Caucasian < Indian)	
Family history of diabetes (offspring and first-degree relatives of patients with type 2 diabetes mellitus)	
Genetic predispositions	
Other risk factors	
Gestational diabetes ^a	
Tuberculosis ^a	
Anti-retroviral therapy ^a	

^aNot well studied.

used measure of obesity, although other anthropometric parameters, such as waist circumference and waist-to-hip ratio, are also used [51]. In SSA, obesity has not been much of a concern until recently [52]. The low levels of obesity may be explained by the undernourishment and low levels of urbanization and mechanization causing most of the populations to lead largely physically active lifestyles in the past [53]. As per the data compiled by Martorell *et al.* (2000), the percentage of female obese subjects within the 15–49 age group in 16 countries in SSA was relatively low (1.0–7.1%), as compared to that in some of the leading economies, where >20.0% of the subjects were reported to be obese [54]. However, data published in 2002 showed that South Africa already had staggering combined figures for overweight and obesity ($\text{BMI} \geq 25 \text{ kg/m}^2$) in the adult population, being 56.6 and 29.2% for women and men, respectively [55]. Both studies indicated that some of the key determinants of obesity in SSA are age, education, ethnicity, and urbanization [54,55].

Studies have shown obesity to be an important risk factor for developing T2DM [56], along with other contributing factors of sedentary lifestyle and the intake of high-energy foods. Studies in Mali, Tanzania, and Nigeria showed higher prevalence of T2DM with increases in BMI [15,17,21,41]. Also, several other studies indicated a higher BMI in diabetic patients compared to that in non-diabetic individuals [17,29,35,41]; a similar trend was observed with obesity [28,29,35,41]. BMI and obesity have also been specifically assessed, and each was found to be an independent risk factor for T2DM [21,29,32,36,41,43].

It has recently been suggested that abdominal obesity is a more important risk factor for T2DM and other metabolic complications associated with obesity than subcutaneous obesity [51]. Therefore, assessments of waist circumference and waist-to-hip ratio in addition to BMI should be more relevant. However, there is debate on the need to re-adjust the anthropometric measures for SSA from the set conventional WHO cutoffs [57], as the prevalence of T2DM is common even in lean subjects, unlike in the case of American whites and blacks [58]. A study carried out in urban Cameroon showed that abdominal obesity was present in 67% of women and 18% of men [59], a pattern similar to that found in most developing nations [48]. A more recent study by Christensen *et al.* (2008) of three ethnic populations in Kenya showed higher mean visceral than subcutaneous fat by a factor of 4.6–11.9 in both genders [60]. Also, a notable pattern in developing countries is the emerging prevalence of overweight and abdominal obesity among children, who are then placed at risk for T2DM in adulthood [11,48,61]. However, overweight and obesity prevalence in SSA children is lower than in their Western counterparts, owing to the prevalence of stunting and malnutrition [11,61]; eradication of these two issues also requires more attention. The lower overweight and obesity rates among the children should not, however, be interpreted to mean that these issues do not require any

attention, as some country-specific overweight and obesity trends, mainly in the urban regions of SSA, have shown a markedly rising pattern [11,62]. In addition, because studies now suggest that intrauterine growth retardation and the resulting low birth weight predispose individuals to metabolic disorders in their adulthood [44,63,64], the high prevalence of stunting and malnutrition [11,61] in SSA might escalate the development of these disorders. This combination of childhood stunting and adulthood obesity has led to the proposition that such individuals in their early years acquire the contentious 'thrifty phenotype' such that when they rapidly adapt to sedentary lifestyles and high-calorie diets later in their lives, they will be highly prone to obesity-related disorders [53,63]. The thrifty phenotype hypothesis seems to apply better for the sub-Saharan populations rather than the thrifty gene hypothesis (theorizes that genes derived from times of deprivation may result in adaptations that have adverse effects in times of plenty) [63,65]. This is because the prevalence of T2DM and obesity has been on the increase since the advent of dramatic nutritional/lifestyle changes. The rising trends of urbanization and improvement of SES in SSA [48] will also possibly increase the prevalence of obesity due to the transitional impacts of nutrition and lifestyles. In relation to this, something that cannot be ignored as a contributing factor to obesity is the cultural perception of a large body size, especially within the black African female population. A large body size is viewed positively, as a sign of affluence, good health, attractiveness to the opposite sex, and happiness; it is also well perceived by most men [10,66]. This connotation has had even greater importance since the onset of the HIV/AIDS scourge within the region. Weight loss, wasting, and lipodystrophy are common among patients with late-stage AIDS [67]; the positive perception of a large body size may increase the stigma of a slender body type and also contribute to the spread of HIV (as individuals with large body sizes may be thought to be infection free).

Transitions: nutrition and lifestyle

In SSA, several studies have shown a higher prevalence of T2DM in urban than rural areas [12,22,25,29,36,43], which is attributable to nutritional and lifestyle changes. Furthermore, a study in South Africa found that urbanization was a significant risk factor for T2DM [32]. There has been a shift from the healthy traditional diet in urban areas to diets high in refined carbohydrates, saturated fats, and cholesterol, and low in fibre [14,48]. This diet change, coupled with reduced physical activity associated with sedentary urban lifestyles (a shift from the intense energy-burning agricultural chores and walking to motorized transportation, manufacturing and service industries, indoor entertainment, and mechanization) [14], has increased the risks of developing obesity and T2DM. However, assessing nutritional and lifestyle risks in SSA has been difficult because of methodological limitations, such as cultural differences, classification

of food groups, and imprecise collection of information [68]. Despite these shortcomings on nutrition and lifestyle assessment, studies focusing on food habits need to be undertaken in SSA given that a recent study by Napoli *et al.* [69] in Cameroon showed unexpectedly high rates of obesity and dysglycaemia among the rural population. This is contrary to the belief that this is the case in the urban settings. This trend has also been observed by Popkin [70] in other developing nations, suggesting a global change in food environments that initially starts in urban regions and possibly spills to the rural centres at a greater pace. This study by Popkin [70] also indicated that the dramatic nutritional changes included increased intakes of edible oils, animal foods, and caloric sweeteners with lower fibre intake that are mainly driven by global increases in food supply and decreases in large-scale pricing. On the other hand, Sobngwi *et al.* (2002) found a negative correlation between physical activity and fasting blood glucose in urban males [71]. T2DM has also been attributed to physical inactivity [72]; therefore, nutritional and lifestyle changes are strong independent risk factors for T2DM. Also, with the total population in African urban areas by 2025 being projected to be 51.7% (752 million), up from 37.8% (310 million) in 2000 [48], this rapid urbanization pattern will increase exposure to these risks. Other lifestyle changes of note are alcohol consumption and cigarette smoking, which have been reported to moderately increase T2DM risk [68] but remain to be well studied in SSA. However, a few studies have correlated frequent alcohol consumption with increased glucose intolerance and T2DM, mainly among men [31,73]. Smoking prevalence has been reported by Townsend *et al.* (2006) to be higher in males (30–49 age group) from 14 SSA countries than in females, whose rates increased steadily with age [74]; so this trend should be investigated in association with T2DM and its comorbidities.

Transitions: socioeconomic

Previous studies in SSA have reported a high prevalence of adiposity [54,59,61] to be strongly associated with populations of a high SES, as indicated by income, education levels, and urbanization. However, it has been argued that this pattern is no longer tenable in SSA [48,75–77] and other developing countries as gross domestic product has now increased and low SES populations now access sufficient food and adopt relatively physically inactive lifestyles, especially those who move from rural to urban areas. Furthermore, alcohol consumption and smoking have been reported to be common in populations of low SES [78]. Foreign direct investment targeted at developing countries, including those in SSA, is increasing, mainly from transnational food companies, which has made highly processed foods available to many people of both high and low SES because of the low prices [79]. Similarly, the increasing availability of cheap edible oils and fats in SSA [80] might

also be driving their higher consumption, along with deep-fried foods that are rich in trans fats. Therefore, it is necessary that the impacts of nutritional transitions be given more attention by transnational food companies and that nutrition intervention programs geared to address undernourishment in SSA be carefully designed.

Transitions: demographic

Ageing, a key demographic transition indicator, has been reported to be a significant risk factor for T2DM in SSA [29,32,43]. As noted before, a high prevalence of T2DM was found in the 45–64 age group [12,21,28,35,43], and in some instances, the >65 group [12,15,17,23,29,32,36]. A higher prevalence within middle-aged populations compromises their productivity and predisposes them to developing T2DM complications in the long term, thus impacting negatively the economic and health care systems. However, the negative impacts of the HIV/AIDS pandemic and other communicable diseases on life expectancy cannot be underestimated, as this might reduce the number of people at risk for T2DM. Mechanization, another demographic indicator, is increasing in urban areas [48] and might also spill over to rural areas as socioeconomic development progresses, thus compromising the rural, physically active lifestyles.

Other risk factors

The genetic contribution to T2DM risk has been supported by studies in identical twins [68]. However, this contribution is multifactorial, as numerous diabetogenic alleles working in association appear to be required, and one or even a few cannot fully explain the genetic susceptibility to T2DM [68]. In SSA, T2DM prevalence has been shown to vary with ethnicity [12], suggesting a certain degree of genetic predisposition. In addition, family history was found to be an independent risk factor for T2DM [24,29,43] but not in a study in South Africa [32]. A recent report from the ongoing Africa American Diabetes Mellitus study on the genome-wide search for susceptibility genes for T2DM identified four major genetic loci (10q23, 4p15, 15q14, and 18p11) influencing C-peptide concentration in West Africans with T2DM [81].

The association of TB and diabetes in SSA has been recently reviewed by Young *et al.* (2009), who suggest that just as diabetes increases the risk for TB, the converse might also be true [82]. This proposal is supported by previous findings that presence of TB could lead to the development of diabetes because most of the TB patients had impaired glucose tolerance (IGT) [83,84]. However, these studies do not provide definite underlying mechanisms, and so further detailed examination is needed. The increasing use of anti-retroviral therapy for HIV/AIDS is also projected to cause adverse metabolic abnormalities among SSA patients [82]. Also, whereas the

occurrence of gestational diabetes has been documented in SSA [85], data is lacking on the relation of this gestational diabetes to the later onset of T2DM among these subjects and their infants.

Pathogenesis

One of the major causes of T2DM is progressive β -cell dysfunction in the presence of chronic IR [86]. IR is a subnormal biological response to a given concentration of insulin [87] in adipose, muscle, and liver cells [68]. This, for instance, results in the overproduction of glucose by the liver despite fasting hyperinsulinaemia and hyperglycaemia [87]. It has been suggested that adipose tissues play a role in the development of inflammation and the elevation of plasma free fatty acids (FFAs) in obese subjects, which appear to contribute to IR [68]. The high levels of circulating FFAs lead to the increased entry of FFAs into non-adipose cells that have limited fat storage capacity, causing lipotoxicity and, in the case of β -cells, probably in combination with hyperglycaemia (glucotoxicity), reductions in function and mass [68]. Genetic and environmental factors such as physical inactivity also play important roles in the pathogenesis of T2DM. Therefore, T2DM is a rather complex and multifactorial metabolic disease [68,88].

IR is strongly indicated to be an etiological factor in T2DM, but this has been disputed [86]. Clinical data have shown that almost 80% of type 2 diabetic patients are insulin resistant and that obese subjects with or without T2DM are mostly insulin resistant [86,89]. IR also appears to precede and cause β -cell dysfunction through β -cell exhaustion [86,90,91]. In SSA, the few studies that have assessed IR among patients with type 2 diabetes strongly support the presence of reduced insulin sensitivity [58,92–95], which is also manifested in their non-diabetic first-degree relatives [96,97]. Race and ethnicity have been found to play a role in determining insulin sensitivity, with black Africans and Asian Indians being less insulin sensitive than whites in SSA [98–103]. Interestingly, Goedecke *et al.* (2009) recently showed a higher acute insulin response to glucose among the more insulin-resistant, normoglycaemic blacks than among whites of South Africa [103], thus presenting a possible explanation for their normoglycaemic state. Altogether, IR is a common feature in patients with type 2 diabetes and high-risk subjects, and so it is likely an etiological factor in T2DM in SSA.

Although IR is highly correlated to T2DM, the individuals who do not develop T2DM usually compensate for the reduced insulin sensitivity by increasing insulin secretion, thus eliciting normoglycaemia [86,87]. Individuals with IGT have increased fasting glucose and postprandial insulin levels that do not fully compensate for IR [87]. This increases the risk of their developing T2DM because of the failure of the β -cell compensatory mechanism at this stage, which will lead to further deficiencies in insulin

secretion and to the exacerbation of hyperglycaemia [86–88]. The underlying mechanisms for the β -cell failure have been suggested to be hereditary and acquired, such as those stemming from nutrient excess (glucotoxicity and lipotoxicity) [87]. The prevalence of IGT in SSA has been increasing with T2DM [41], and the current IDF atlas (2009) estimates a prevalence of 1.1–14.1% for IGT by the end of 2010 in SSA, with Ghana being ranked among the top ten countries worldwide in prevalence (14.1%) [6]. The increasing trend in IGT prevalence is therefore expected to enhance the rising trend of T2DM and CVD because IGT is highly associated with these disorders [104]. As with other populations, SSA individuals with IGT and T2DM have been reported to manifest blunted acute first insulin secretion in response to intravenous or oral glucose challenge [58,94,97,105,106]. Osei *et al.* (2003) also showed that insulin and C-peptide responses are significantly lower during both oral and intravenous glucose challenges in type 2 diabetic patients when compared to healthy non-diabetic native Ghanaians [58]. Also, in the same study, patients with IGT showed slightly higher total serum insulin and C-peptide responses, which occurred at the expense of higher postprandial serum glucose levels compared to the healthy non-diabetic subjects [58]. Similarly, non-diabetic first-degree relatives and the offspring of type 2 diabetic patients in Nigeria and Cameroon showed marked reductions in early-phase insulin secretion [96,107]. However, this was not the case with first-degree relatives of type 2 diabetic patients in Ghana, who, instead, had manifestations of hyperinsulinaemia after both oral and intravenous glucose challenges when compared to healthy controls; this result might have been due to differences in anthropometric parameters and other exogenous factors [108]. From another standpoint, black non-diabetic obese Africans in South Africa were found to have lower secretory levels of both insulin and C-peptide than whites after oral glucose tolerance challenge followed by tolbutamide–glucagon stimulation [106]. der Merwe *et al.* (1999) also found lower fasting insulin and C-peptide levels in blacks than in whites in both non-diabetic/obese and diabetic women [109]. In addition to this, the progression of β -cell failure in the black South African diabetic patients was found to be more rapid than in whites [94]. Therefore, these findings in SSA, though most were limited and cross-sectional, show that β -cell dysfunction is critical to the pathogenesis of T2DM, in which genetic and acquired factors might also be involved and cause varied effects. In retrospect, these β -cell dysfunctions in SSA type 2 diabetic patients seem to agree with the hypotheses of Weir and Bonner-Weir [110] on stages of β -cell dysfunction during progression to diabetes. These stages include β -cell compensation in the face of IR, stable β -cell adaptation, and IGT followed by decompensation, respectively.

As noted earlier, adipose tissue might also play a significant role in the pathogenesis of T2DM [68], which has been an area of interest for researchers lately. Adipose tissue, together with the macrophages that infiltrate this tissue during obesity [111], secrete

a number of adipokines, such as leptin, adiponectin, tumour necrosis factor- α , and FFAs; these factors promote inflammation and IR [112]. Leptin is associated positively with increases in BMI and adiposity, increased insulin sensitivity, and regulation of appetite/energy homeostasis [112]. However, increased leptin levels in obesity do not lead to improved tissue sensitivity to insulin because of the development of leptin resistance [109]. In SSA, a study by der Merwe *et al.* (1999) found higher leptin and FFA concentrations and lower insulin levels in two groups of black South African women, obese/non-diabetic and obese/diabetic patients, compared to their white counterparts [109]. The higher FFA levels that have also been documented previously in obese black women compared to those in whites [113] might be due to higher lipolytic activity as a result of resistance to the anti-lipolytic effect of insulin more than literally decreased insulin secretion levels [114]. This resistance to anti-lipolytic effect of insulin, together with the effects of leptin resistance [109], might exacerbate the FFA concentrations, causing IR and lipotoxicity. On the other hand, the concentration of adiponectin decreases with increasing adiposity, unlike other adipokines, precluding its beneficial effect on insulin sensitivity and prevention of vascular injury [101,115,116]. Investigations of adiponectin levels among type 2 diabetics in SSA are scant, but there have been reports of lower levels of adiponectin in black Africans and Asian Indians when compared to whites, with different correlates, such as leptin, triglycerides, and IR [117,118], suggesting a possible ethnic difference in adiponectin's mode of action. Biomarkers of inflammation, such as C-reactive protein, have also been correlated with obesity, IR, and T2DM within SSA populations [119] in addition to being a strong predictor of CVD risk. CVD development has been associated with dyslipidaemia [9], another condition now reported within SSA diabetic patients [120]. Future studies will therefore be pivotal to understanding these emerging concepts in the pathogenesis of T2DM in SSA.

Morbidity and mortality

Poor glycaemic and blood pressure control may lead to the various specific chronic vascular complications that precede the onset of T2DM in SSA patients [121]. Microvascular complications, e.g. retinopathy, neuropathy, and nephropathy, are reported to be far more common than macrovascular diseases, e.g. stroke, coronary artery disease, ischemic heart disease, and peripheral vascular disease (PVD), despite the strong association between hypertension and diabetes in SSA [14]. This report contrasts with those from most developed nations, where macrovascular diseases are the leading cause of both morbidity and mortality among type 2 diabetic patients [8]. At diagnosis, retinopathy already affects 21–25% of type 2 diabetes patients in SSA [122,123], and the reported prevalence of diabetic

retinopathy lies between 16 and 55% [121]. It was reported that the black Africans were more prone to T2DM than Indians and Caucasians in SSA [122]. Although genetic predisposition may account for the differences among different ethnic groups, hypertension and poor glycaemic control appear to be significant precipitating factors [121]. In developed countries, diabetic retinopathy is the leading cause of adult blindness [124], but this association is not well assessed in SSA. However, macular edema or ischaemia and/or proliferative retinopathy manifest in 15% of SSA patients with severe retinopathy [14,125,126], and the susceptibility to cataracts and glaucoma have also been reported to be more prevalent in diabetic patients than in healthy populations [127–129]. Similarly, nephropathy occurs early in the course of diabetes, usually in association with retinopathy [130]; 32–72% of patients with a mean duration of diabetes of 5–10 years were reported to have microalbuminuria [121,125,126,131], while overt proteinuria was found in 5–28% of the patients and increases with diabetes duration [121,125,132]. On the other hand, neuropathy interacts with PVD and infection to cause foot ulcerations [121,133,134], leading to amputation when sepsis occurs [134]. In SSA, the majority of diabetes-related foot ulcers (>80%) are neuropathic rather than ischemic [135]. The prevalence of peripheral neuropathy varies between 10 and 36% [121], and erectile dysfunction has been reported in up to 49% of diabetic patients [132,136–138].

Although the occurrence of macrovascular diseases is relatively uncommon in SSA compared to that in industrialized countries, there is now an increasing evidence of a rising prevalence within type 2 diabetic patients and/or as part of the metabolic syndrome [139–141]. Recently, Steyn *et al.* [142], through the INTERHEART Africa study group, also documented five major modifiable CVD risk factors (diabetes, hypertension, smoking, abdominal obesity, and ratio of apolipoprotein B to apolipoprotein A-I) that are attributable to 89.2% of the acute myocardial infarction cases found in nine SSA countries. Coronary artery disease has been reported to affect 5–8% [132,138,141,143] of type 2 diabetes patients and 50% of those with cardiomyopathy [141]. On the other hand, up to 5% of type 2 diabetic patients in SSA have manifestations of stroke at diagnosis [138,141,144]. When the absence of a foot pulse was used as the detection method for PVD, PVD prevalence ranged between 4 and 8%, compared with 18–28% when Doppler ultrasound examinations were performed [10,125,141,145–148]. Altogether, reported vascular complications among SSA type 2 diabetic patients are limited and appear to have wider prevalence ranges as a result of differences in diabetes duration, glycaemic control, classical CVD risk factors, ethnicity, and possibly diagnostic facilities [135,141].

Renal failure and CVD are the main causes of death among T2DM patients in Western countries, but infection and acute metabolic complications are reported to be the main causes of mortality in SSA [128]. These causes of mortality among T2DM patients in SSA were

reported from outcome studies of T2DM in Zimbabwe [149], Ethiopia [150,151], and Tanzania [152] that indicated mortality rates from T2DM of 8–41% [41]. Acute complications include diabetic ketoacidosis, often due to insulin deficiency or delayed diagnosis [153], and hyperosmolar non-ketotic coma [154,155], while hypoglycaemia occurred in some patients who were on anti-diabetic drugs [156]. Foot sepsis [132,157], TB [158], and occasionally mucormycosis and hand infections [159,160] are among the reported common infections. However, with the increasing trend of vascular complications, most of which require expensive medical care, such as dialysis and transplantation for renal failure [161], compounded by increases in disease duration, these vascular complications will probably emerge as key causes of mortality among T2DM patients. It is worth noting that the total mortality from T2DM or all diabetes in SSA is not well documented and remains a difficult task. This lack of data is because causes of death on death certificates may indicate underlying complications (rather than diabetes), and because some out-of-hospital deaths are likely not reported [10].

Treatment

The importance of glycaemic control in the reduction of diabetes-related vascular morbidity and mortality in both type 1 diabetes mellitus and T2DM was strongly demonstrated by the Diabetes Control and Complications Trial [162] and the United Kingdom Prospective Diabetes Study [163]. Long-term glycaemic control is usually determined through measurements of glycosylated haemoglobin levels in diabetic patients; levels of 7% or less are usually desirable [8]. Lifestyle modifications and diet interventions should be part of the therapy for all T2DM patients; these changes benefit not only glycaemic control but also other CVD risk factors [164]. Truter (1998) investigated the anti-diabetic medications widely prescribed in South Africa and found that most (64.4%) patients received only oral hypoglycaemic agents (OHAs), so it was assumed that most of the patients had type 2 diabetes [165]. Half of these patients exclusively on OHA were using sulphonylureas (SUs) and a third were on SUs in combination with a biguanide. The most prescribed OHA was the SU gliclazide (38.2%), followed by the biguanide metformin (33.3%). On the other hand, 30.5% of all the patients received only insulin, and 5.2% had insulin and one or more OHA. However, a recent study by Yusuff *et al.* (2008) among type 2 diabetic patients in Nigeria indicated that there is indeed a shift from OHA monotherapy to combination therapy with at least two OHA agents, often from different classes with or without insulin [166], which might promote better glycaemic control [167]. Specifically, Yusuff *et al.* (2008) found that OHA was only prescribed to 86% of the cohorts (171), while an insulin and OHA combination were prescribed

to 14% (29). Glibenclamide, an SU, and metformins were the most prescribed OHA combination and were the dominant oral therapy prescribed, but hypoglycaemia was documented in 60.3% of the patients. Poor glycaemic control was also documented by Otieno *et al.* (2008) in type 2 diabetics under different treatments (insulin only, OHA only or OHA with insulin) and with glycosylated haemoglobin levels of 8.07–9.32% [168]. This poor glycaemic control in SSA patients under treatment can be partly ascribed to the limited access to treatment regimens due to the cost, short supply, late diagnosis, poor adherence to prescription, and unsuccessful self-management by patients [166]. However, poor glycaemic control is not only evidenced among SSA patients; it has been reported to remain a problem worldwide, regardless of the setting or anti-diabetic therapy used [167]. Side-effects of therapy, such as weight gain, and the fact that the therapies cannot correct the β -cell dysfunction lead to long-term glycaemic problems and consequent increases in vascular complication risks [8]. Treatment of these comorbidities may in itself promote the deterioration of glycaemic control; e.g. the use of β -blockers among hypertensive patients can increase adiposity [169]. Alternately, failure to treat these complications could also challenge the achievement of glycaemic control. Otieno *et al.* (2005) reported that dyslipidaemia and hypertension were untreated and/or inadequately controlled among type 2 diabetics attending an outpatient clinic in Kenya [170]. Recently, Justin-Temu *et al.* (2009) reported that the first-generation SUs chlorpropamide and tolbutamide were the only OHA agents, along with insulin, stocked in three public facilities in Dar es Salaam, Tanzania, while the private facilities had a variety of OHA agents as well as insulin [171]. However, the costs of the drugs in the private facilities were over five times higher compared to that in public facilities, and so affordability limits flexibility in obtaining the optimal use of anti-diabetic therapies. Overall, the currently validated core therapies (lifestyle intervention, metformin, SUs, and insulin) [167] appear to have been adopted in SSA, while the less validated therapies, such as α -glucosidase inhibitors, thiazolidinediones, glinides, dipeptidyl-peptidase-4 inhibitors, glucagon-like peptide-1, and amylin agonists, are rarely used.

Care challenges and their solutions

It is apparent that SSA faces problems in attaining optimal care and management of T2DM, mainly due to the strained economic resources in the region [172]. Not surprisingly, limited resources are allocated to the governmental health ministries, which directly affects NCD management because communicable diseases receive more attention [173]. The annual cost of diabetes (1989–1990) was estimated at \$103 per diabetic patient not requiring insulin in Tanzania, whereas the allocation of funds for health expenditures was \$2 per capita

Table 2. Key challenges to diabetes care and suggested solutions in sub-Saharan Africa

Challenges to diabetes care ^a	Solutions to diabetes care problems ^b
Poor patient attendance	Reorganization of health care system
Poor patient education	Prioritization of diabetes prevention
Inadequately trained nurses and health care workers	Expansion of data collection
Limited training programs for health care staff	Provision of diagnostic tools and infrastructure
Unsystematic evaluation and monitoring of diabetes complications	Improvement of drug procurement and supply
Poor glycaemic and blood pressure control among patients	Advocacy for accessibility and affordability of medicines and care
Inadequate health referral systems	Improvement of training and availability of health care workers
Poor organization in service delivery to patients	Enhancing patient adherence to care
Lack of a structured record keeping plan for patients and in health care facilities	Increasing patient education and empowerment
	Aggressive integration of community and diabetes associations
	Ensuring a positive policy environment

^aWhiting *et al.* [178].^bBeran *et al.* [173].

during that year [174]. Global funding also largely supports communicable disease management programs [173]. With high biomedical costs compounded by sociocultural perceptions, some patients prefer traditional healers or use both traditional and medical treatments [10,175,176], thus reducing timely disease diagnosis and prescription adherence. SSA patients face not only an issue of affordability of medication but also of availability. For instance, insulin availability has been interrupted at some time in most developing countries [173]. A more recent study by the International Insulin Foundation also showed that insulin availability still remains a problem in Mali, Zambia, and Mozambique [177], and the case is likely to be similar in most other SSA countries. This situation was investigated using the Rapid Assessment Protocol for Insulin Access, developed by the foundation that assesses the paths of insulin and other diabetes-related supplies and identifies barriers to care. The key challenges to diabetes care in Africa have also been reviewed in depth by Whiting *et al.* [178] and are mainly due to poor patient attendance and education as well as an inadequate health care system among other reasons, as summarized in Table 2.

The high cost of treating T2DM and its comorbidities, the increasing prevalence of its modifiable risk factors, and the gaps in health care systems in resource-poor settings such as SSA no doubt provide strong and valid reasons why interventions/solutions have to be implemented. These solutions must involve patients, families, communities, governments, health

systems, academia, and organizations/associations. The strategies should be cost-effective if well planned [42]. Solutions will be achieved by solving known problems; therefore, some of the key points to be addressed for positive diabetes care in SSA have been identified by the International Insulin Foundation and authored by Beran and Yudkin [173] and mainly involve various organizational aspects in health care, education, and policy making structures (Table 2).

Despite the gaps in medical care and the management of diabetes in SSA, a few evidence-based interventions, mainly initiated by local hospitals and external funding sources, have been relatively successful [135]. Also, efforts now being spearheaded by the IDF, International Insulin Foundation, WHO, Diabetes Action Now, Diabetes UK, Novo Nordisk, World Diabetes Foundation, Santé Diabète Mali, Health Action International, local Diabetes Associations, Ministries of Health, etc., appear to provide promising opportunities for achieving workable NCD infrastructure in SSA [173]. Notably, the launch of the Declaration of Diabetes and Strategy for Africa in 2006 through partnership efforts of IDF, WHO, and Africa Union set out a call and plan of action to prevent and control diabetes and related chronic illness [179]. Perhaps, the most notable initiative that has been implemented successfully at a national level is the national diabetes and hypertension program in Cameroon [180]. This program has been backed up by research data in a positive political environment, despite the involvement of multisectoral partnerships. In the long term, the same model is planned for the surveillance of other NCDs in the country such as cancer, epilepsy, sickle cell disease, and stroke. Therefore, aggressive actions and positive responses from well-informed governments are largely needed for the interplay of all forces required to curb the growing threat of T2DM in SSA.

Conclusion

The increasing prevalence of T2DM in SSA will negatively impact the health care system and socioeconomic development in the region. The burdens of T2DM are multiple, since its comorbidities, e.g. vascular diseases, complicate the management of this disease. Health care systems in SSA are inadequate and more structured towards the management of the communicable diseases, which are still a major problem in this resource-poor region. It is therefore important to initiate cost-effective intervention programs geared at reducing the prevalence of the modifiable risk factors, such as obesity and physical inactivity, which can be ascribed to rapid urbanization and socioeconomic transitions. At the same time, improvement/and or expansion of health care systems should be targeted to achieve optimal care of patients with T2DM and its related chronic disorders along with the communicable diseases. Several initiatives from both local and international organizations present

promising opportunities for solving the challenges facing T2DM patients and expanding the limited existing data from this region. However, awareness initiatives on the threats of T2DM and its related chronic illness to the SSA populations and execution of plans of action need to be urgently implemented extensively.

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Conflict of interest

None declared.

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